RAG · INTERVIEW PREP · 2026

Top 20 RAG interview questions

Every AI engineer should know these. Fundamentals, architecture, ingestion & chunking — in plain English. 👇

Fundamentals | Architecture | Ingestion | Chunking

Ashwini
Senior AI/ML Engineer

01 · FUNDAMENTALS

What is Retrieval-Augmented Generation (RAG)?

RAG lets a language model look up information before it answers.

Without RAG — a student answering from memory alone

With RAG — the student opens a book, reads the right page, then answers

Up-to-date, private-data-aware answers with fewer made-up facts.

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02 · FUNDAMENTALS

Limits of a pure LLM (no RAG)?

A pure LLM can only rely on what it already knows:

- 🧊 **Frozen knowledge** — trained last year → doesn't know yesterday
- 🔒 **No private data** — can't see policies, internal docs, records
- 🌀 **Hallucination** — confidently wrong answers that sound right

General knowledge, old information, higher risk of errors.

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03 · FUNDAMENTALS

RAG vs. Fine-tuning

Fine-Tuning

❌ Changes the model itself

❌ Knowledge baked in

❌ Slow, costly to update

❌ Focus: behavior & style

RAG

✅ Model stays the same

✅ Knowledge lives outside

✅ Fast, cheap to update

✅ Focus: knowledge access

RAG = reference books in the exam. Fine-tuning = teaching new subjects.

04 · FUNDAMENTALS

When prefer RAG over fine-tuning?

Reach for RAG when your data is:

- **Changing often**
- **Large**
- **Private / company-specific**
- **Needs transparency**

FINE-TUNE INSTEAD WHEN

you need to change tone, teach special rules, or specialize on one task. Many real systems use RAG first, fine-tuning only if needed.

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05 · FUNDAMENTALS

What problems & data suit RAG?

Best for knowledge problems where the answer already exists in documents:

- Doc Q&A
- Enterprise search
- Legal systems
- Tech docs assistants

📚 **Sources** — PDFs, Word, text, web pages, DBs, wikis, chat logs

🚫 **Less useful** — creative writing, pure reasoning puzzles

Docs are chunked, embedded, and stored in a vector database.

The two core components of RAG?

Every RAG app is built on two halves working together:

🔍 **Retrieval** — searches external sources for relevant information

✍️ **Generation** — an LLM combines the question + retrieved context into an answer

Goal: factual accuracy meets natural-language generation.

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07 · ARCHITECTURE

How does indexing work — and why?

Indexing prepares documents so they can be searched fast:

1 · **Split** — break documents into small chunks

2 · **Embed** — convert each chunk into a numeric embedding

3 · **Store** — save embeddings in a vector database

Without it, searching large collections is slow and inaccurate.

08 · ARCHITECTURE

The full RAG pipeline

❓ User query

↓

🔢 Query embedding

↓

🔍 Retrieve relevant chunks

↓

🏆 Rank & select

↓

🧩 Build context

↓

💬 Generate answer

> The LLM reads the actual document instead of guessing from memory.

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09 · DATA INGESTION

What do DocumentLoaders do?

They bring content from anywhere into one standard format:

📥 **DocumentLoaders** — web, PDFs, DBs → LangChain Document (text + metadata)

🌐 **WebBaseLoader** — downloads a URL's HTML, extracts readable text

✂️ **+ TextSplitter** — chunks that raw text — retrieval works best small

Load in a standard shape, then split before embedding.

10 · DATA INGESTION

Why customize HTML parsing?

RAG quality depends on what you store in the vector DB:

🍜 **BeautifulSoup** — parses HTML tags, selects useful parts, extracts clean text

🎯 **SoupStrainer** — parses only specific parts (e.g. <article>) — faster, cleaner

THE STAKES

Store noise like "Cookie settings" or menus and the LLM answers poorly. Clean parsing → only real content → accurate retrieval.

11 · CHUNKING

Why split documents into chunks?

Large documents mix many topics into one blurry vector:

- 🎯 **One idea per chunk** — each chunk focuses on a single topic
- 📐 **Token limits** — embedding models can't take unlimited text
- 🔍 **Precise search** — one paragraph = sharp meaning, better similarity

Kept whole, embeddings go vague and retrieval gets inaccurate.

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Your turn.

Which RAG question trips you up most?

Drop it in the comments and I'll answer. Save this for your next interview, and follow for more AI engineering prep in plain English.

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